

Radar Signal Sorting via Graph Convolutional Network and Semi-Supervised Learning

Ziying Li , Xiongjun Fu , Senior Member, IEEE, Jian Dong , and Min Xie

Abstract—As a key technology in radar reconnaissance systems, radar signal sorting aims to separate multiple radar pulses from an interleaved pulse stream. Supervised signal sorting methods based on deep learning depend on a large volume of training data to optimize model parameters. However, acquiring labeled pulses in practice is challenging. In this letter, a semi-supervised learning (SSL) framework is proposed to address this issue. First, a Self-Organizing Map (SOM) is used to learn the spatial distribution of impulse features, and an anchor graph is constructed based on SOM nodes. A pseudo-label set is then generated using the SOM based on pulse discrepancy information. Finally, a three-layer Weighted Residual Graph Convolutional Network (WRGCN) is designed for signal sorting, with its parameters pre-trained on pseudo-labels and fine-tuned with a limited number of true labels. Experiments on a simulated radar pulse dataset demonstrate that this framework outperforms several existing methods for radar signal sorting with limited labeled pulses.

Index Terms—Radar signal sorting, graph convolutional network, pseudo-labels, semi-supervised learning.

I. INTRODUCTION

RADAR signal sorting (RSS) is an integral component of the radar electronic reconnaissance system [1]. Using RSS, radar receivers can distinguish signals emitted from various radiation source within interleaved and overlapping radar pulse sequences [2], [3]. The main parameters utilized in RSS constitute the Pulse Description Word (PDW), which includes Direction of Arrival (DOA), Time of Arrival (TOA), Pulse Width (PW), Radio Frequency (RF), and Pulse Amplitude (PA).

Early sorting methods primarily relied on PRI, such as Cumulative Difference Histogram (CDIF) [4], Sequential Difference Histogram (SDIF) [5], and PRI Transformation Method [6], [7]. However, PRI-based methods are highly susceptible to PRI errors, making it difficult to achieve stable signal sorting. Consequently, research has shifted towards clustering methods [8] based on PDW, which mainly includes K-Means clustering [9], [10], spatial density clustering [11], [12], fuzzy clustering [13], [14], etc. While these algorithms can achieve relatively good sorting performance under certain conditions, they tend

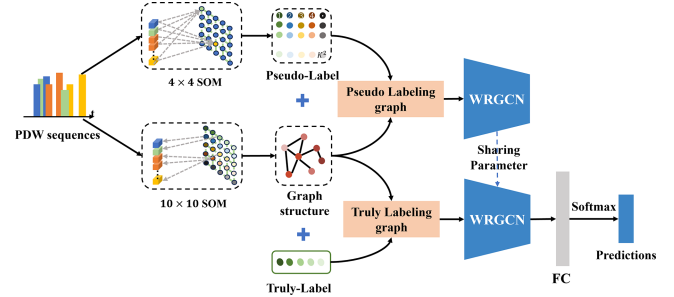


Fig. 1. Overall framework of the proposed method.

to perform poorly when faced with complex radar feature sets [9]. Traditional methods are likely to incorrectly classify modes from one transmitter into different transmitters, leading to the problem of “increasing batch” [15].

In recent years, rapid advancements in deep learning have introduced new solutions for radar signal sorting. By framing radar signal sorting as an image recognition or time series classification problem, deep learning techniques from these domains can be effectively applied to achieve accurate signal sorting [16], [17], [18]. For example, Wu et al. [16] utilized the regularity of PA sequences to train a neural network prediction model based on gated recurrent units (GRUs). However, these networks need a substantial amount of correctly labeled samples for training, restricting the application of such methods in scenarios with limited labelled pulses.

Semi-supervised learning (SSL) [19], which utilizes limited labeled data alongside large amounts of unlabeled data to enhance performance, has garnered significant attention. SSL methods include semi-supervised clustering algorithms [20], semi-supervised support vector machines [21], and graph-based SSL approaches [22], etc. Among them, graph convolutional network (GCN) [23], [24], [25], [26], which considers both node features and graph structure, has achieved success in semi-supervised node classification [27], [28]. Anchor graph is a graph-based learning model for large datasets [28] [29]. It constructs a novel graph by using data points and a few anchor points, leveraging their linear relationship. Compared to KNN graph, it reduces the computational complexity on graph construction from $O(N^2)$ to $O(NM)$. (N is the number of data points and M is the number of anchor points) [30].

Given that the PDW can be treated as graph nodes and organized into a KNN graph for node classification [32], we propose a semi-supervised learning (SSL) framework comprising the following three stages, as illustrated in Fig. 1. Firstly, the spatial distribution of pulse features is learned using self-organizing map (SOM), and an anchor graph structure is constructed based

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on SOM nodes that represent target patterns. Next, utilizing the competitive learning mechanism of SOM, the winning SOM nodes are employed to label the input data in reverse order, thereby generating a pseudo-label set. Finally, all the training data is used to pre-train the proposed WRGCN, which is then fine-tuned using the limited labeled data.

The main contributions of this paper are as follows:

1) A SSL framework is developed that is well-suited for radar signal sorting with limited labeled prior information. (2) A lightweight anchor graph construction method based on SOM is proposed, which effectively characterizes the graph structure of radar pulse sequences. (3) A pseudo-label generation method based on SOM is proposed that effectively exploits hidden label information in unsupervised clustering. (4) An improved GCN structure, WRGCN, is constructed and a SSL method based on pseudo-labels is proposed, which further enhances network generalization performance with fewer labeled samples.

II. PROPOSED METHOD

A. Problem Formulation

Given a set of radar pulse sequences $X = (x_1, \dots, x_i, \dots, x_N)$, $X \in \mathbb{R}^{N \times d}$ where $x_i \in \mathbb{R}^{d \times 1}$ is the feature vector of each pulse. N is the number of radar pulses and d is feature dimension. $D^l = \{(X_i^l, Y_i^l)\}_{i=1}^{N_l}$ and $D^u = \{X_j^u\}_{j=1}^{N_u}$ denote labelled and unlabelled radar pulses where Y_i^l is the label of X_i^l , $N_l \ll N_u$.

An undirected graph can be represented by $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. $\mathcal{V} = \{v_1, v_2, v_3, \dots, v_N\}$ is the set of nodes. $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges, while the adjacency matrix $A \in \mathbb{R}^{N \times N}$ represents the relationships between nodes. Radar pulse sequences X can be represented as a graph, where each pulse feature vector corresponds to a node in the graph. The deinterleaving process can then be viewed as a graph node classification task. The proposed method is using D^l and D^u to train a node-level classifier, obtaining a label set of X_j^u .

B. SOM-Based Graph Construction of Radar Signals

To effectively explore the relationships among various types of interleaved signals, we use SOM to construct a graph of the radar pulses. SOM [33] is an unsupervised learning method that maps high-dimensional space X into a discrete 2-D grid $M = \{m_k\}_{k=1}^{K^2}$, where m_k denotes the centroid vector of node k . A competitive learning strategy is employed to train the SOM. After initializing the centroid vectors of $K \times K$ nodes, the following steps are repeated until all feature vectors in X are input:

- 1) Randomly select an input vector $\{x_i\}_{i=1}^N \in X$. Calculate the distance between the input vector and all nodes according to (1). Identify the node closest to the input feature vector x_i , as the 'winning' node c .

$$c = \arg \min_k \|x_i - m_j\|, j = 1, 2, \dots, K^2 \quad (1)$$

- 2) During the process of weight adaptation, the centroid vectors of the 'winning' node and its neighboring nodes are updated as follows:

$$m_j(t+1) = m_j(t) + \eta(t)\pi_{i,c}(t)(x_i(t) - m_j(t)),$$

$$j = 1, 2, \dots, K^2 \quad (2)$$

$$\pi_{i,c}(t) = \exp\left(-\frac{\|x_i - m_c\|^2}{g(t)}\right) \quad (3)$$

Where $\eta(t)$ is the learning rate, and $\pi_{i,c}(t)$ is the topological neighborhood.

After training, the centroid set C of SOM winning nodes is then used as the anchor points to construct the anchor graph, fitting the feature space of radar pulses. A Gaussian kernel function $K(x_i, c_j)$ is employed to compute the similarity matrix between radar pulses and anchor points as follows:

$$K(x_i, c_j) = \exp\left(-\frac{\|x_i - c_j\|_2^2}{2\sigma^2}\right) \quad (4)$$

$$w_{ij} = \frac{K(x_i, c_j)}{\sum_{l=1}^{K^2} K(x_i, c_l)} \quad (5)$$

Where σ represents the parameter controlling the width of the neighborhood.

To construct the complete graph, the adjacency matrix A is defined as:

$$A = W\Lambda^{-1}W^T \quad (6)$$

$$\Lambda_{jj} = \sum_{i=1}^N w_{ij} \quad (7)$$

This approach fully utilizes the SOM's ability to capture the topological structure of input feature space. The extracted anchor points are more distributed and representative. Compared with the KNN-based graph construction method, it is more suitable for graph construction of radar pulse sequences.

C. Generation of Pseudo-Labels Via SOM

Pseudo-labels refer to the artificial labels generated by the model itself or the cluster assignment attributes generated by some clustering algorithms [34]. In order to make full use of the prior difference information of the unlabelled samples, SOM described in Section II-B is also used for pre-sorting to generate pseudo-labels.

After the unsupervised training of SOM converges, samples are re-input sequentially. For each new input sample, its distance to the centroid vectors of $K \times K$ nodes is calculated, with the closest distance determining the best matching node. Then, each matching node is labelled, and the input samples are labelled in reverse according to their best matching nodes' label, as shown in Fig. 2. The generated pseudo-labels Z indicate that input samples corresponding to the same matching nodes belong to the same class.

D. Weighted Residual Graph Convolutional Network (WRGCN)

To enhance the performance of GCN, a three-layer graph network model called WRGCN is proposed. Specifically, the propagation of the proposed model includes three steps:

- 1) Normalization: the adjacency matrix of the anchor graph A is normalized according to $A_S = D^{-1/2}AD^{-1/2}$, where D is a degree matrix of graph, $d_{jj} = \sum_{i=1}^n a_{ij}$.

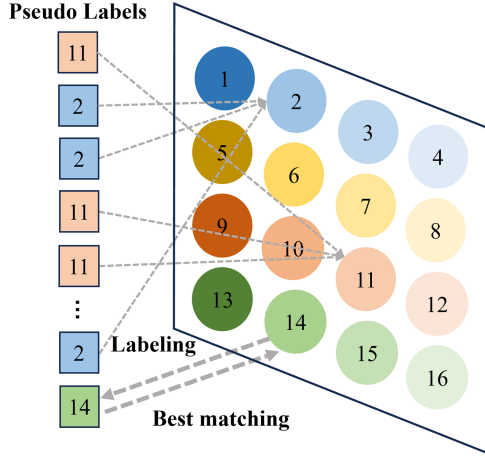


Fig. 2. Pseudo-label generation process of SOM.

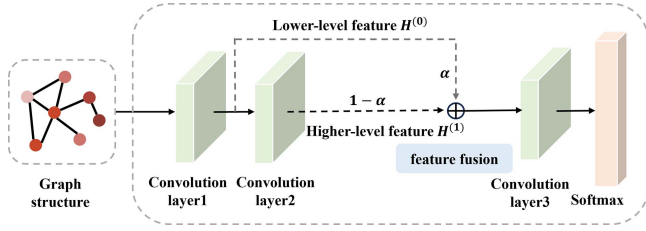


Fig. 3. Illustration of the WRGCN module.

- 2) Propagation: the layer-wise propagation takes the following form:

$$H^{(k+1)} = \sigma(A_S H^{(k)} \Theta^{(k)}) \quad (8)$$

where $H^{(k+1)}$ denotes the $(k+1)$ layer. $\Theta^{(k)}$ is the trainable weight matrix in layer and σ is the activation function.

- 3) Prediction: three layers of graph convolutional networks are stacked, and a Softmax function is applied to the output features to generate the prediction matrix.

$$F = \text{Softmax} \left\{ \tilde{W}_s \sigma \left(\tilde{W}_s \sigma \left(\tilde{W}_s H^{(0)} \Theta^{(0)} \right) \Theta^{(1)} \right) \Theta^{(2)} \right\} \quad (9)$$

We introduce a skip connection from the first layer, directly integrating first-order neighborhood aggregation information into the second layer. Additionally, a weighting parameter is introduced to adjust the contribution of different input components.

$$H^{(2)} = \sigma(A_S((1-\alpha)H^{(1)} + \alpha H^{(0)})\Theta^{(k)}) \quad (10)$$

The final output can be interpreted as a label matrix, and the cross-entropy loss is used to penalize any deviation between the output and the labels of the training data.

$$\mathcal{L} = - \sum_{i=1}^{N_l} \sum_{j=1}^d \mathbf{X}_{ij}^{\text{out}} \ln(\mathbf{Y}_{ij}) \quad (11)$$

Where N_l is the index of nodes with labels, $\mathbf{X}^{\text{out}}$ is the output, and $\mathbf{Y}$ is the label matrix. The gradient descent algorithm uses the entire training set in each iteration. The structure of the WRGCN is shown in Fig. 3.

TABLE I
PARAMETER SETTINGS FOR RADAR EMITTER

	MFR	PW/ μs	RF/GHz	DOA/ $^\circ$	PRI/ μs	Pulse number
No.1	I	195~200	8.0~9.8	36.8~37.2	800 Fixed	250
	I	90~140	8.3~8.5	35.5~36.5	400~500 Sliding	444
No.2	II	115~120	8.4~9.5	35.5~36.5	450~550 Jittered	400
	III	130~160	8.8~9.4	35.5~36.5	600 Fixed	333
No.3	I	95~100	8.0~9.2	35.0~36.0	500 Fixed	400
	I	130~160	9.2~9.5	36.0~37.0	300/330/370 Jagged	600
No.4	II	150~170	9.5~9.6	36.0~37.0	400 Fixed	500
	III	160~200	9.6~9.8	36.0~37.0	500~700 Sliding	333

E. Semi-Supervised Deep Learning Using Pseudo Labels

As discussed in Section III-C and III-D, pseudo-labels Z can be used to train the WRGCN, enabling the hidden layers to retain the discriminative information from pre-sorting. However, pseudo-labels still contain misclassified radar pulses. It is necessary to address the ‘increasing batch’ generated during pre-sorting by batch merging.

Firstly, pre-training the WRGCN using pseudo-labels, where the number of neurons in the third layer is equal to the number of pseudo-label categories. Then, all network parameters are frozen, and two fully connected layers and a classification layer (i.e., Softmax classifier) are added after the third graph convolutional layer. Fine-tuning is performed using a limited number of labeled samples. Only the parameters of the added fully connected layers are updated.

III. EXPERIMENTS AND RESULTS

A. Datasets Description

Four MFR individuals are simulated in our experiments, each with one to three operating modes. DOA, PW, and RF are used as the typical characteristics of each radar pulse, which can exhibit various modulation types, including conventional, slip, parametric, and jitter. The same parameter types are overlapped with varying degrees of validity between different radars or operating modes. For example, for the two operating modes of Radar 2, the DOA variation range is the same, and the overlap rates of PW and RF are 40% and 70%, respectively. For Radar 2 and Radar 3, the overlap rates of RF and DOA are 64.29% and 33.33%, respectively. The detailed information of the simulations is shown in Table I.

B. Experimental Setup and Results

To verify the effectiveness of our method, the performance of all methods is evaluated under the labelling rate of 10%.

In the graph construction stage, the SOM grid size is set to be 10×10 , while for pseudo-label generation, the SOM grid size is set to be 4×4 . In the pre-training of the WRGCN, the whole sample is used as the training set, with 10% used as validation set. The pre-training learning rate is 0.05. The architecture of the WRGCN (from the input layer to the output layer) is chosen as 16-16-16. The weight factor α determining the proportion of

TABLE II
PER-CLASS ACCURACY, OA, AA, AND KC OF DIFFERENT METHODS WITH LABELING RATE OF 10%

Methods	Radar1	Radar2	Radar3	Radar4	AA	OA	KC	T/s
PSO-K [9]	80.40%	45.37%	99.60%	71.18%	74.13%	66.10%	55.75%	14.11
ABC_K [10]	88.40%	44.35%	98.25%	82.07%	78.26%	70.92%	60.95%	3.54
LPA [22]	59.46%	58.54%	33.43%	100%	62.85%	74.00%	55.32%	0.44
SVM [21]	86.67%	85.55%	62.91%	98.99%	83.53%	88.73%	82.45%	0.31
CS-K [20]	94.00%	73.41%	99.50%	98.05%	91.24%	89.02%	84.60%	0.94
ResGCN-RSS [32]	95.45%	91.78%	94.68%	95.03%	94.23%	94.30%	92.12%	8.91
Proposed SSL Framework	99.52%	96.76%	94.69%	95.39%	96.58%	95.67%	94.09%	10.83

feature fusion is chosen as 0.3. After pretraining, two randomly initialized fully connected layers are added, with dimensions chosen as 64–64. The learning rate of fine-tuning is 0.005. The experiments are carried out in a Linux system, on an Intel(R) Xeon(R) Gold 5118 CPU @ 2.30G processor computer.

We compare the proposed method with other methods, including unsupervised methods PSO-K [9] and ABC_K [10], and semi-supervised methods SVM (with linear kernel) [21], Constrained Seed K-Means (CS-K) [20] and Label propagation Algorithm (LPA) [22], ResGCN-RSS [32]. The overall accuracy (OA), the average accuracy (AA), and the kappa coefficient (KC) were adopted as the evaluation metrics.

Table II presents the experimental results, demonstrating that our proposed framework outperforms all the previous methods in terms of OA, AA, and KC. Specifically, PSO-K and ABC_K show poor overall accuracy, with results of just 66.10% and 70.92%, respectively. Compared to three semi-supervised learning methods: LPA, SVM, and CS-K, our framework achieves overall accuracy improvements of 21.67%, 6.94%, and 6.65%, respectively, highlighting the advantages of using graph structure and graph network for radar signal sorting. Compared to ResGCN-RSS, our framework achieves higher accuracy, with improvements of 4.07% and 4.98% for Radar 1 and Radar 2, respectively. At the same labeling rate of 10%, the overall sorting accuracy reaches 95.67%, marking an improvement of 1.37%. Although the proposed method takes slightly longer than ResGCN-RSS, the use of anchor graphs significantly reduces computational complexity during the graph construction phase, rendering the small increase in time cost acceptable.

As shown in the Fig. 4(a) and Fig. 4(b), the sorting method based on K-Means classifies Radar 2, which has two operational modes, into two radiation sources, resulting in “increasing batch”. DBSCAN ultimately classifies the 4 classes of radar into only 2 classes of radar. As shown in the Fig. 4(c), SVM still has the problem that Radar 1 is classified as Radar 4, and Radar 2 is classified as Radar 3. Our proposed framework, as shown in the Fig. 4(d), achieves nearly correct classification for all 4 radars.

C. Performance on Small Labeled Data

To validate the effectiveness of our framework in data-constrained scenarios, we compared it with LPA, SVM, CS-K, and ResGCN-RSS at various labeled data rates. As shown in Fig. 5, our framework consistently outperforms the others. At a 1% labeled data rate, it achieves over 88.51% accuracy, compared to 53.86% for LPA, 73.89% for SVM, 85.62% for CS-K, and 86.55% for ResGCN-RSS. Performance improves as the labeling rate increases, reaching about 95% accuracy when the rate exceeds 5%, highlighting the advantages of our framework in low-labeling scenarios.

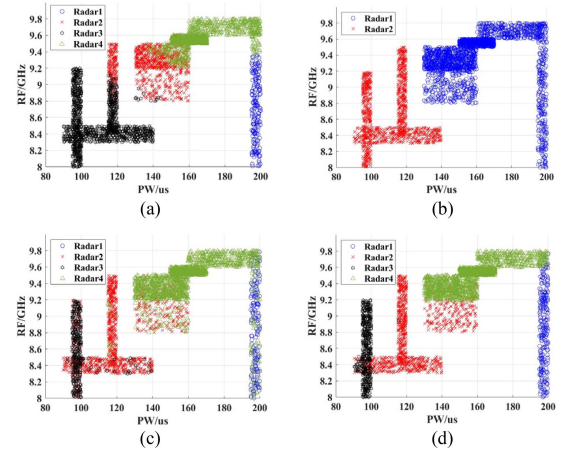


Fig. 4. Results of different methods. (a) K-Means. (b) DBSCAN. (c) SVM. (d) The proposed SSL framework.

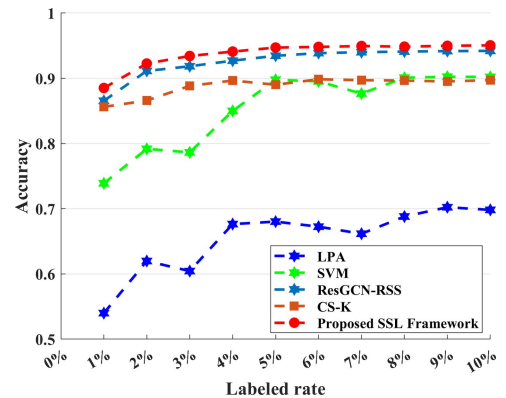


Fig. 5. Sorting accuracy comparison with different data labelled rate.

IV. CONCLUSION

This letter focuses on the problem of semi-supervised radar signal sorting. A SSL framework is proposed to address the challenge of acquiring a large quantity of labelled radar pulses in real-world scenarios. Comprehensive experiments are conducted on a simulated radar pulse dataset with label proportions ranging from 1% to 10% to evaluate sorting performance. The results demonstrate that our proposed method achieves higher accuracy compared to several existing unsupervised and semi-supervised algorithms. In future work, we aim to further explore more complex and challenging radar signal sorting problems using semi-supervised learning.

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